

Basic Tasks

This is a collection of basic tasks, you can complete all of them using only variables, conditional operators and loops.

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Hello world

Just print Hello world phrase.

Output Format

string value.

Standard output

Examples

```
Hello world
```

Find greatest

You are given 3 numbers, a b c. Find the greatest one.

Input Format

a, b, c: integer values. Each one on a new line.

Standard input

Output Format

integer value.

Standard output

Examples

```
5
2
3
```

```
5
```

Even number

Check if given number n is even.

Input Format

n : integer value.

Standard input

Output Format

boolean value.

Standard output

Examples

3

false

100

True

Comparing

You are given two numbers a and b . If a is greater than b print "A", otherwise print "B".

Input Format

a, b : integer values. In one line, separated by whitespace.

Standard input

Output Format

boolean value.

Standard output

Examples

```
9 7
```

```
A
```

```
-100 100
```

```
B
```

You can try to do it without directly checking $a > b$, but comparing something with zero. That is how it happens on the lowest level.

Area

You are given coordinates of rectangle vertices. Calculate the area of this rectangle.

Input Format

Four pairs of floating point values. Starting from upper-left corner, clockwise.

Standard input

Output Format

Area of rectangle : floating point value.

Standard output

Examples

```
2.2 3.3  
5.2 3.3  
5.2 -1.2  
2.2 -1.2
```

```
13.5
```

Looping

Print number 1 ten times.

Output Format

a sequence of integer values, separated by whitespace.
Standard output

Examples

1 1 1 1 1 1 1 1 1 1

Breaking the infinity

Run an infinite loop. Inside it generate a random value between 0 and 1000. Your task is to count the amount of iterations.

But, if generated value is divisible by 3 - we don't count it as iteration. And if generated value squared is bigger than 100000 we end the loop.

Output Format

integer value.

Standard output

Examples

Lets say we generated this: 100 98 33 2 2 90 1000

5

Too many loops

Print all possible three number combinations of numbers from 1 to 4. Lets say combination is good if the sum of its numbers equals to a given value n. Count the amount of good combinations.

Input Format

n : integer value.

Standard input

Output Format

A list of combinations, each one on a new line.

Count of good combinations : integer.

Standard output

Fibonacci

Print n-th number of fibonacci sequence (1 1 2 3 5 8 13 21)

Input Format

n : integer value.

Standard input

Output Format

n-th number of sequence : integer value.

Standard output

Examples

4

2

10

55

Vowels

Count amount of vowels in given string s. (vowels: a i u e o)

Input Format

s: string value.
Standard input

Output Format

integer value.
Standard output

Examples

hello world

3

there are 11 vowels in this sentence

11

Replace All

You are given a string `s` consists only of symbols “@”, “\$” and “#”.
Replace all symbols in a string, which count is minimal by symbols,
which count is greatest

Input Format

`s` : string value.

Standard input

Output Format

string value.

Standard output

Examples

```
##@@@$$$$
```

```
@@@@@$$$$
```

```
$$$$$$@$#####
```

```
$$$$$$##$#####
```

Sequence*

There is a sequence: 1,1,2,1,2,3,1,2,3,4... and so on.
Print out first N numbers of this sequence.

Input Format

N : integer value.
Standard input

Output Format

N first numbers of the sequence.
Standard output

Examples

3

1 1 2

7

1 1 2 1 2 3 1